

Chi-square Tests

Hypothesis Testing: Using chi-square tests (χ^2)

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Where We Left Off

If you have a continuous/interval DV and...

- ...a dichotomous IV:

Where We Left Off

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- ...a dichotomous IV: use a t-test!
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- ...a dichotomous IV: use a t-test!
- ...a nominal/ordinal DV: use an ANOVA!

ANOVAs:

- ANOVAs help us compare the means of *multiple* groups within a variable
- They're called "analysis of variance" tests, but they test the means (they assume equal variances)

What is the ANOVA testing, exactly?

- H_0 : the difference in means between groups is 0 (i.e., there is no significant difference in means)
- H_1 : the difference in means between groups is greater/less than 0 (i.e., there *is* a significant difference in means)

But Wait, There's More!

Remember: An ANOVA test only answers the question of whether there is a significant difference in the mean of the DV across *all* categories of the IV.

But what if the mean of the DV is significantly different for some IV categories but not others?

We can find that out using an additional test: **Tukey HSD!**

Revisiting Cross-tabulations

Crosstabs show how the dependent variable is distributed across different values of the independent variable (IV).

But we use chi-square to *test* expected relationships among the data organized in crosstabs.

What are the ground rules for crosstabs again?

Revisiting Cross-tabulations

Crosstabs show how the dependent variable is distributed across different values of the independent variable (IV).

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What are the ground rules for crosstabs again?

- Rows: DV categories
- Columns: IV categories
- **Always** calculate the column percentages
- Interpretation goes across rows (at the same value of the DV), then down columns.

Crosstab example

Favorite color (DV) by Tolerance for bullcrap (IV)

	Low	Medium	High	TOTAL
Green	756 30.9%	233 59.7%	170 40.4%	1159 35.6%
Blue	858 35.1%	106 27.2%	151 35.9%	1115 34.3%
Red	829 33.9%	51 13.1%	100 23.8%	980 30.1%
TOTAL	2443 100%	390 100%	421 100%	3254 100%

Notice where they come from: take the cell value divided by the total column value at the bottom (e.g. $756/2443 = .309$ then times 100 for percentage).

Chi-square tests

Chi-square tests are useful when we have both of the following:

- Dichotomous, Ordinal, or Nominal IV
- Dichotomous, Ordinal, or Nominal DV

Chi (pronounced “kai”) is a Greek letter (χ).

Chi-square helps us test for “independence” among variables to see if our crosstab depicts a statistically significant relationship.

What is chi-square testing, exactly?

- H_0 : There is no relationship between IV and DV
- H_1 : There is a relationship between IV and DV (we can even be more specific about that relationship with real variables).

Let's Try It

Let's pretend a couple of fictional students named Schason and Scherek got into an argument about whether the relationship between bullcrap tolerance and favorite color is statistically significant...

Schason: Ugh, this is just a ridiculous example. There's no way there's a relationship.

Scherek: The data doesn't lie, dude! It looks like it could be significant!

Schason: You're full of it. I won't tolerate this bullcrap. I'm ROTC.

Scherek: Is your favorite color blue, by any chance?

Schmackenzie, probably: Why are you *still* arguing? That's embarrassing for you. We can just do a Chi-square test and find out...

Is there a significant relationship between one's favorite color and their tolerance for bullcrap?
Let's find out.

The Math (Just for Funsies)

Don't sweat the math steps to follow; they just illustrate what's going on with t-tests in the background.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

The χ^2 statistic is equal to the sum ($\sum$) of each of the observed cell values (O_i) minus the expected cell values (E_i) squared, over the expected cell values.

In other words, Chi-square is examining how different the cells are from what we would expect to see in regularly distributed data.

But what do we *expect* exactly?

Calculating Expected Values

For each cell, look at the other values in the row, and the other values in the column. Let's start with the top left (756)...

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$$E_i = \frac{(\text{Sum of row}) * (\text{Sum of column})}{\text{Grand total}}$$

$$E_i = \frac{1159 * 2443}{3254} = \mathbf{870.1}$$

Some Subtraction and Averages

Name	O_i	E_i	$(O_i - E_i)$	$(O_i - E_i)^2$	$(O_i - E_i)^2/E_i$
G-L	756	870.1	-114.1	13018.81	14.96
G-M	233	138.9	94.1	8854.81	63.75
G-H	170	150.0	20	400	2.67
B-L	858	837.1	20.9	436.81	0.52
B-M	106	133.6	-27.6	761.76	5.70
B-H	151	144.3	6.7	44.89	0.31
R-L	829	735.8	93.2	8686.24	11.81
R-M	51	117.5	-66.5	4422.25	37.64
R-H	100	126.8	-26.8	718.24	5.66
χ^2					143.02

So, our calculated chi-statistic (χ) for comparison is 143.02. Now, we can compare it to the critical value of χ^* .

Finding the Critical Value

Chi-square (χ^2) tests are called that because they rely on the χ -distribution.

Our degrees of freedom is $(r - 1) * (c - 1)$ (we have 3 rows and 3 columns, so 2x2 is 4 df!) and we want a significance level of 0.05 (remember 95% confidence and $p < 0.05$!)

Chi-square Distribution Table

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22

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9.49

Is there a significant relationship?

Yep. $\chi > \chi^*$

To see whether or not there is a significant relationship between bullcrap tolerance and favorite color, we can compare our calculated χ -statistic to the critical value of χ^* :

Because our χ -statistic (143.02) is greater than our critical χ^* value (9.49)...

- we **reject** the null hypothesis.
- we can conclude that there **is a statistically significant** relationship between bullcrap tolerance and favorite color.

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OR we could let R do all this and find the p-value of a quick chi-square test...